GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005051

Subject Name: Fertilizer Technology

w.e.f. Academic Year:	2026-27
Semester:	V
Category of the Course:	PEC-04

Prerequisite:	Foundational knowledge of Fertilizer Technology, unit operations, and unit processes is required.
Rationale:	Indian economy is dominated by agriculture sector. Synthetic fertilizers are must for producing good crops. Hence it is needed to provide comprehensive and balanced understanding of essential link between chemistry and the synthetic fertilizer industry. It is therefore vital for chemical engineers to understand for each fertilizer product, its flow diagram for Industry production. For this purpose chemical engineers should have skills for arranging treatment, reaction and separation steps in a flow diagram for variety of fertilizers including Nitrogenous fertilizers, Phosphatic fertilizer, Potash Fertilizer, Complex fertilizer and Bio fertilizers is essential. Hence this course is designed to achieve this objective.

Course Outcome:

After Completion of the Course, Student will able to:

No.	Course Outcomes	RBT Level
01	Explain the classification and properties of various fertilizers.	R/U
02	Apply concept of the manufacturing processes of various fertilizers with neat sketches to operate chemical plant.	A
03	Identify major engineering problems encountered in manufacturing processes.	R/U
04	Suggest applications of various fertilizers.	R/U/A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+(PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial/Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	Marks	% of Weightage
Unit-I (Overview of Fertilizer)	1.1 Define fertilizer and explain need of it 1.2 Classification of fertilizers 1.3 Role of essential Elements in plant Growth 1.3.1 Macro elements 1.3.2 Micro elements 1.4 Application of fertilizers 1.5 Calculate nutrient content in given fertilizers	04	6	8%
Unit-II (Nitrogenous Fertilizers)	2.1 Physical, chemical properties and applications of ammonia, nitric acid and urea 2.2 Manufacturing of ammonia synthesis gas 2.3 Explain sketch of ammonia synthesis converters 2.4 Manufacturing of ammonia 2.5 Storage and Transportation of Ammonia 2.6 Manufacturing of Nitric Acid 2.7 Concentration of Nitric acid by $Mg(NO_3)_2$ 2.8 Manufacturing of Urea 2.9 Major engineering problems of ammonia, nitric acid and urea manufacturing 2.10 Manufacturing of 2.10.1 Ammonium nitrate 2.10.2 Ammonium sulphate 2.10.3 Ammonium chloride	15	22	33%
Unit-III (Phosphatic Fertilizer)	3.1 Physical, chemical properties and applications of elemental phosphorus and phosphoric acid 3.3 Manufacturing of elemental phosphorus 3.3 Manufacturing phosphoric acid by Wet Process 3.3.1 Strong Sulphuric Acid Leaching 3.3.2 Hydrochloric Acid Leaching 3.3.3 Electric Furnace Process 3.4. Major engineering problems of phosphorus and phosphoric acid 3.5. Manufacturing of Single Superphosphate 3.6 Manufacturing of Triple superphosphate	09	14	20%
Unit-IV (Potassic Fertilizers)	4.1 List out various potassic fertilizer 4.2 Physical, chemical properties, manufacturing and uses of 4.2.1Potassium chloride	06	10	14%

	4.2.2 Potassium nitrate 4.2.3 Potassium sulphate 4.2.4 Potassium magnesium sulphate (K_2SO_4 $MgSO_4$) 4.2.5 Potassium carbonate (K_2CO_3)			
Unit-V **(Complex Fertilizer)**	5.1 Define complex fertilizer and list out it 5.2 Manufacturing of 5.2.1 NPK fertilizer 5.2.2 Ammonium Sulphate Phosphate (ASP), 5.2.3 Calcium Ammonium Nitrate(CAN) 5.2.4 Di-ammonium phosphate 5.2.5 Nitrophosphate	05	10	11%
Unit -VI **(Bio Fertilizer)**	6.1 Define Biofertilizers and classify it 6.2 Biofertilizers: 6.2.1 Nitrogen-fixing 6.2.2 Phosphate solubilizing 6.2.3 Potassium solubilizing 6.3 Preparation of a biofertilizers 6.4 Classify organic fertilizers with examples 6.5 Outline advantages of organic fertilizers over Conventional fertilizer	06	08	14%
	Total	**45**	**70**	**100%**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in%)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
19	**32**	**19**	**-**	**-**	**-**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (asper Revised Bloom's Taxonomy)

***Note**: This specification table provides general guidelines to assist students for their learning and to teachers to teach and question paper designers/setters to formulate test items/questions to assess the attainment of the UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may slightly vary from above table.

Suggested Course Practical List:

Sr. No.	Practical Outcomes (PrOs)	Unit No.
1	Prepare chart for fertilizer classification with chemical formula and nutrient content	1
2	Estimate nutrient content (% N, %P_2O, % K_2O) in different fertilizers from their chemical formula	1
3	Estimate percentage of Nitrogen in Ammonium chloride by substitution method	2
4	Estimate percentage of Nitrogen in Ammonium sulfate by substitution method	2
5	Estimate percentage of Nitrogen in Ammonium chloride by back titration	2
6	Estimate percentage of Nitrogen in Ammonium sulphate by back titration	2
7	Analysis of Urea by Formaldehyde method	2
8	Estimate percentage of Nitrogen in Ammonium Chloride /Sulphate by Kjeldhal's method	2
9	Estimate biuret content in Urea sample by colour comparison	2
10	Prepare potassium sulphate	4
11	Prepare potassium chloride	4
12	Prepare potassium nitrate	4
13	Estimate ratio from Ammonia to Phosphoric acid in DAP	5
14	Estimate percentage of Nitrogen in DAP by Formaldehyde method	5
15	Estimate percentage of Nitrogen in DAP by Kjeldhal's method	5
16	Preparation of Organic fertilizer	6

Note

*i. More **Practical Exercises** can be designed and offered by the respective course teacher to develop the industry-relevant skills/outcomes to match the COs. The above table is only a suggestive list. Course teacher can select any 15 practical.*

*i. The **following are some sample** 'Process' and 'Product' related skills (more may be added/deleted depending on the course) with approximate percentage weightage that occur in the above listed **Practical Exercises** of this **course** required which are embedded in the Cos and ultimately the competency.*

Sr. No.	Sample Performance Indicators for the PrOs	Weightage in% (Approximate)
1	Prepare experimental setup accurately.	10
2	Use apparatus for precise measurements.	20
3	Practice and adapt good and safe measuring techniques.	10
4	Good Record keeping of the observations accurately.	20
5	Interpret the results and their conclusion.	20
6	Prepare Report in prescribed format	10
7	Viva-Voce	10
Total		**100**

Suggested Activities for Students:

Other than the classroom and laboratory learning, the following are the suggested student- related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should perform the following activities in group and prepare small reports of about 5 pages for each activity. They should also collect/record physical evidence such as photographs/videos of the activities for their(student's) portfolio which will be useful for their placement interviews:

a) Prepare a Power Point presentation or animation showing different types of fertilizer manufacturing Process
b) Prepare a model of different fertilizer product flow diagram
c) Preparation of tables showing the difference between Organic and Inorganic fertilizer
d) Market survey of different fertilizer product and compare their physical and chemical properties.
e) Library survey regarding fertilizers in different industries.
f) Collect different fertilizer and prepare the chart/Power Point based on their type, properties, and uses.

References / Suggested Learning Resources:

(a) Books:

Sr. No.	Title of Book	Author	Publication with the place, year and ISBN
1	Outlines of Chemical Technology,3rdedition	M. Gopala Rao, Marshall Sitting	Affiliated East West Press (Pvt) Ltd-New Delhi

2	Shreve's Chemical ProcessIndustries,5th edition	Austin G.T.	McGraw Hill publication–New Delhi
3	Chemical Technology – Vol. I , 2nd edition	G.N. Pandey and Shukla	Vani Books Company -Hyderabad
4	BioFertilizers and Organic Farming	Vayas S C and Modi H A	Akta prakashan, Nadiad

(b) Open sources of software and website:

I. http://nptel.ac.in/courses/103107086/4
II. http://ijset.com/ijset/publication/v1s6/285-291%20IJSET_PK%20JAGA.pdf
III. www.gses.com/images/pressreleases/Manufacturing-Process-Fertilizer.pdf
IV. http://nzic.org.nz/ChemProcesses/production/1A.pdf
V. http://tnau.ac.in/eagri/eagri50/SSAC222/lec12.pdf
VI. www.fnca.mext.go.jp/bf/bfm/pdf/Biofertilizer_Manual.pd

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